GEBZE TEKNİK ÜNİVERSİTESİ

Lojik Devreler ve Tasarım

Bahar 2023

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3. Lojik Kapılar ve Tümdevre Aileleri

Geçen hafta

Sayı Sistemleri:

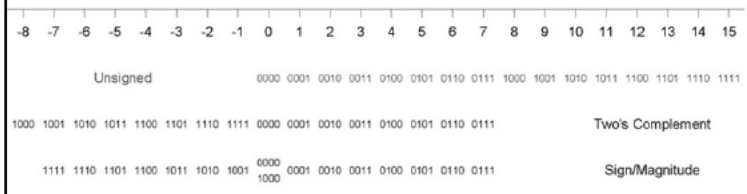
- 2'lik, 10'luk ve 16'lık sayı sistemleri
- Sayı sistemleri arasında geçiş
- 2'lik sistemde toplama işlemi
- Negatif ve Pozitif sayıların gösterimi
  - Sign / Magnitude Method
  - **2's Complement Method**
- İşaret Uzatma (Sign extension)
- Binary Codes (BCD, Gray, ASCII)
  - **Error Detecting Codes, Error Correcting Codes - Okuma**

GEBZE TEKNİK ÜNİVERSİTESİ 2

Review

Number System	Range
Unsigned	$[0, 2^N-1]$
Sign/Magnitude	$[-(2^{N-1}-1), 2^{N-1}-1]$
Two's Complement	$[-2^{N-1}, 2^{N-1}-1]$

For example, 4-bit representation:



Chapter 1 <3>

Boolean Algebra

- **Boolean Algebra** is the algebra of digital logic circuits that all computers use.
- Two possible values: **0 or 1**
- Boolean Variables is often used to represent the voltage level present on a wire or at the input/output terminals of a circuit.

Example:

0V - 0.8V : Logic 0

2V - 5V : Logic 1

Chapter 1 <4>

Logic Level

- Boolean 0 and 1 do not represent actual numbers but instead represent the state of a voltage which is called **a logic level**.

Logic 0	Logic 1
False	True
Off	On
Low	High
No	Yes
Open switch	Closed switch



Truth Table

- A table for describing **how a logic circuit output depends on the logic levels at its inputs.**
- The **inputs** are considered **logic variables** whose logic levels at any time **determine the output levels**.
- Letter symbols are used to represent logic variables for input/outputs.
- Example: 'A' might represent a digital input or output whose value must either be 0 or 1.

A	B	x
0	0	1
0	1	0
1	0	1
1	1	0

A	B	C	x
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1



Examples

- What is the output state of the four-input circuit when all inputs are 1 except A?
- What is the output state if A=1, B=0, C=0, D=1?
- How many table entries needed for a five-input circuit?

A	B	C	D	x
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1



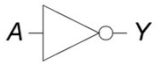
Logic Gates

- Digital circuits called **logic gates** can be constructed from electronics components so that the circuit performs these basic Boolean operations.
- There are *three* basic Boolean algebra operations:
  OR, AND and NOT.



NOT Gate

NOT



$Y = \overline{A}$

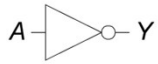
A	Y
0	
1	

- The **overbar** represents the **NOT operation**. It is read as **Y equals NOT A**
- If the input of **NOT** is 1, the output is 0.
- If the input of **NOT** is 0, the output is 1.
- The **NOT** Operation is called inversion, and the digital circuit which inverts is an **inverter**.



NOT Gate

NOT



$Y = \overline{A}$

A	Y
0	1
1	0

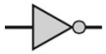
There are a couple notions

- **Y = A'**
- **Y = NOT A**
- **Y = !A**
- **Y = ~A**

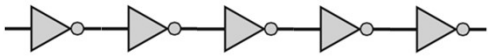


Question

- What is the output logic level of the following inverter with a given logic 0 input?



- What is the output logic level of the following inverter chain connected together with a given logic 0 input?





AND Gate

AND



$Y = AB$

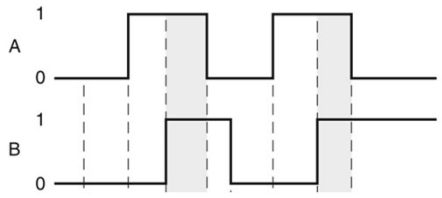
A	B	Y
0	0	
0	1	
1	0	
1	1	

- AND has **two or more inputs**. (Y = A **AND** B **AND** C)
- The output Y of AND, is 1 **if and only if** all the inputs are 1. Otherwise 0.
- It is read as **Y equals A AND B**.
- The · sign might be omitted. So it
- becomes **AB**. It is still the same
- function (A **AND** B)
- Can have more than 2:
  x = ABCD (A **AND** B **AND** C **AND** D)

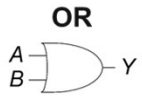


Question

Determine the output Y from the two-input **AND** gate.



OR Gate



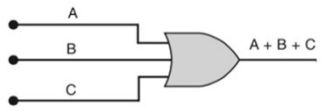
$Y = A + B$

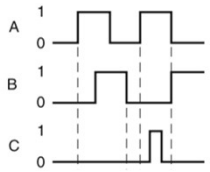
A	B	Y
0	0	
0	1	
1	0	
1	1	

- **OR** has <u>two or more inputs</u>. (Y = A **OR** B **OR** C)
- The output Y of **OR**, is 1 <u>if either one of the inputs are 1</u>.
- It is read as <u>Y equals A **OR** B</u>.
- Can have more than 2 inputs:
  x = A + B + C + D (A **OR** B **OR** C **OR** D)

Question

Determine the output x for the three-input **OR** gate.





AND and OR Gates

AND



$Y = AB$

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

OR



$Y = A + B$

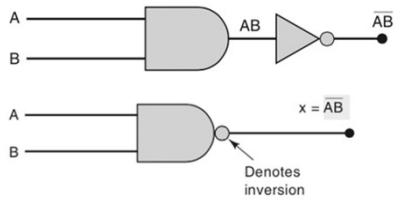
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

NAND Gate

- **Inverted AND**.
- Can have more than 2 inputs.

$$x = \overline{AB}$$

		AND	NAND
A	B	AB	$\overline{AB}$
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

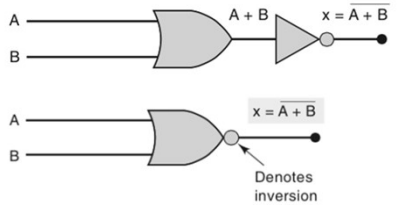


NOR Gate

- **Inverted OR**.
- Can have more than 2 inputs.

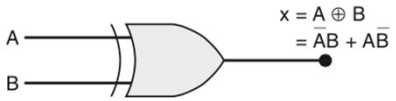
$$x = \overline{A + B}$$

		OR	NOR
A	B	A + B	$\overline{A + B}$
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

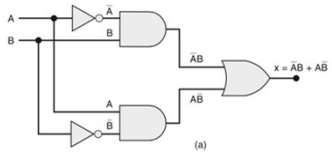


Exclusive-OR (XOR) Gate

- Produces a **HIGH** output whenever the two inputs are at <u>opposite</u> levels.

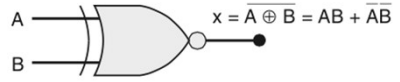


A	B	x
0	0	0
0	1	1
1	0	1
1	1	0

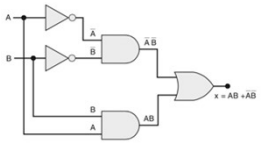


Exclusive-NOR (XNOR) Gate

- Produces a **HIGH** output whenever the two inputs are at <u>same</u> levels.



A	B	x
0	0	1
0	1	0
1	0	0
1	1	1



More Two-Input Logic Gates

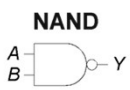
XOR



$Y = A \oplus B$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

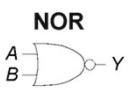
NAND



$Y = \overline{AB}$

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

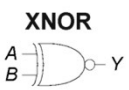
NOR



$Y = \overline{A + B}$

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

XNOR



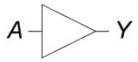
$Y = \overline{A \oplus B}$

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1



Buffer

BUF



$Y = A$

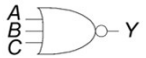
A	Y
0	0
1	1

- Simply copies the input to the output
- Think of it as just a wire



Multiple-Input Logic Gates

NOR3



$Y = \overline{A+B+C}$

A	B	C	Y
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

AND3



$Y = ABC$

A	B	C	Y
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	



Logic Levels

- Discrete voltages represent 1 and 0
- For example:
  - 0 = *ground* (GND) or 0 volts
  - 1 = V_{DD} or 5 volts
- What about 4.99 volts? Is that a 0 or a 1?
- What about 3.2 volts?



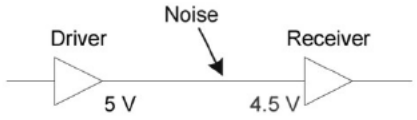
Logic Levels

- *Range* of voltages for 1 and 0
- Different ranges for inputs and outputs to allow for *noise*



What is Noise?

- **Anything that degrades the signal**
  - E.g., resistance, power supply noise, coupling to neighboring wires, etc.
- **Example:** a gate (driver) outputs 5 V but, because of resistance in a long wire, receiver gets 4.5 V



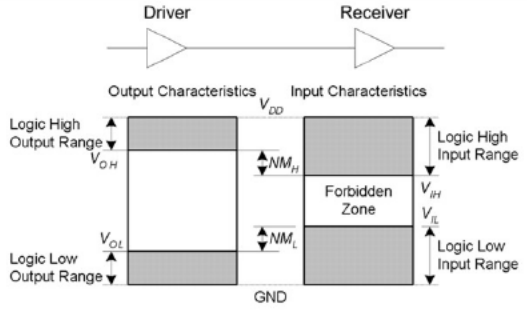


The Static Discipline

- With logically valid inputs, every circuit element must produce logically valid outputs
- Use limited ranges of voltages to represent discrete values

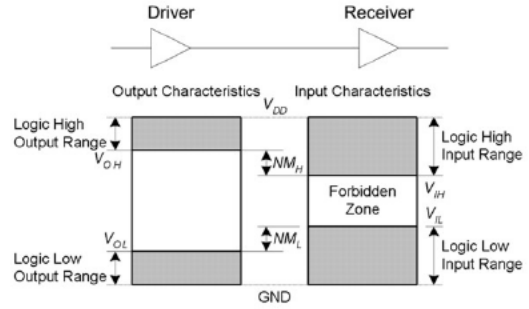


Noise Margins





Noise Margins

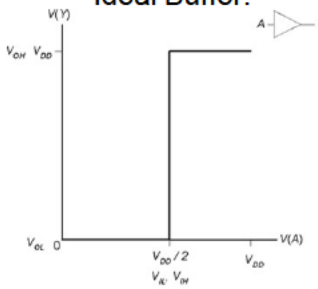


High Noise Margin: $NM_H = V_{OH} - V_{IH}$

Low Noise Margin: $NM_L = V_{IL} - V_{OL}$

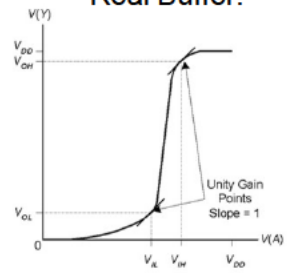
DC Transfer Characteristics

Ideal Buffer:



$NM_H = NM_L = V_{DD}/2$

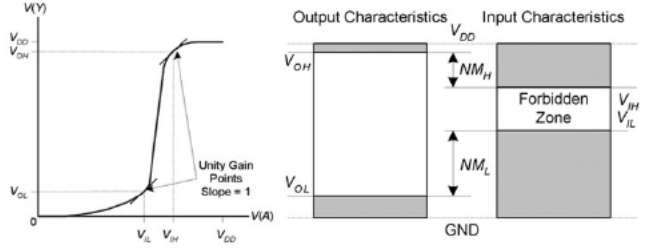
Real Buffer:



NM_H, $NM_L < V_{DD}/2$

DC Transfer Characteristics





V_{DD} Scaling

- In 1970's and 1980's, V_{DD} = 5 V
- V_{DD} has dropped
  - Avoid frying tiny transistors
  - Save power
- 3.3 V, 2.5 V, 1.8 V, 1.5 V, 1.2 V, 1.0 V, …
  - Be careful connecting chips with different supply voltages

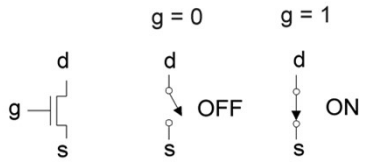
Logic Family Examples

Logic Family	V_{DD}	V_{IL}	V_{IH}	V_{OL}	V_{OH}
TTL	5 (4.75 - 5.25)	0.8	2.0	0.4	2.4
CMOS	5 (4.5 - 6)	1.35	3.15	0.33	3.84
LVTTL	3.3 (3 - 3.6)	0.8	2.0	0.4	2.4
LVCMOS	3.3 (3 - 3.6)	0.9	1.8	0.36	2.7



Transistors

- Logic gates built from transistors
- 3-ported voltage-controlled switch
  - 2 ports connected depending on voltage of 3rd
  - d and s are connected (ON) when g is 1





Robert Noyce, 1927-1990

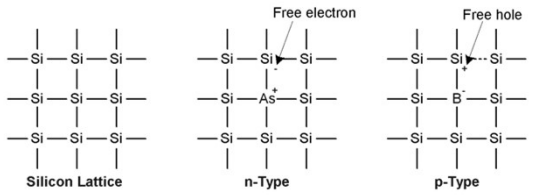
- Nicknamed “Mayor of Silicon Valley”
- Cofounded Fairchild Semiconductor in 1957
- Cofounded Intel in 1968
- Co-invented the integrated circuit





Silicon

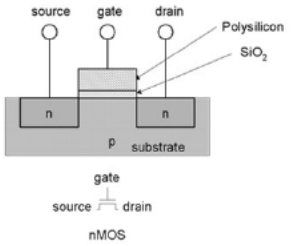
- Transistors built from silicon, a semiconductor
- Pure silicon is a poor conductor (no free charges)
- Doped silicon is a good conductor (free charges)
  - n-type (free ***n***egative charges, electrons)
  - p-type (free ***p***ositive charges, holes)





MOS Transistors

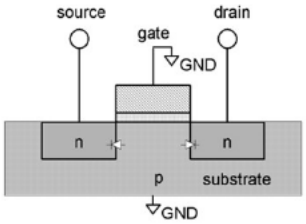
- **Metal oxide silicon (MOS) transistors:**
  - Polysilicon (used to be **metal**) gate
  - **Oxide** (silicon dioxide) insulator
  - Doped **silicon**



Transistors: nMOS

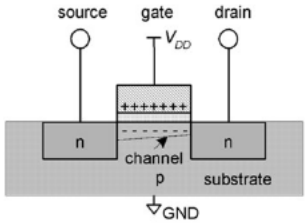
Gate = 0

OFF (no connection between source and drain)



Gate = 1

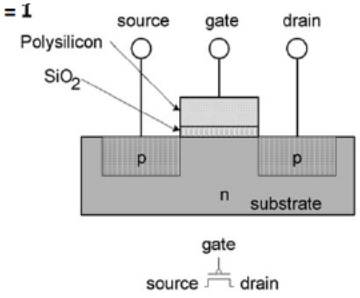
ON (channel between source and drain)



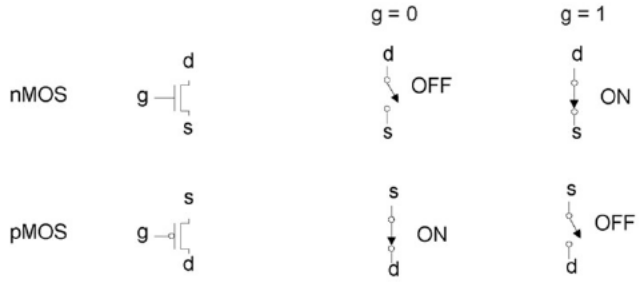
Transistors: pMOS

pMOS transistor is opposite

- **ON** when **Gate = 0**
- **OFF** when **Gate = 1**

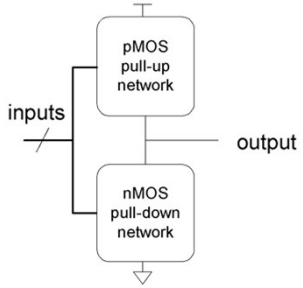


Transistor Function

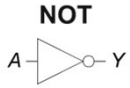


Transistor Function

- **nMOS:** pass good **0**'s, so connect source to GND
- **pMOS:** pass good **1**'s, so connect source to V_{DD}

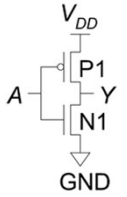


CMOS Gates: NOT Gate



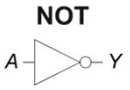
$Y = \overline{A}$

A	Y
0	1
1	0



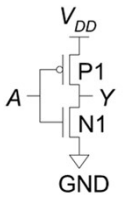
A	P1	N1	*Y*
0			
1			

CMOS Gates: NOT Gate



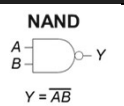
$Y = \overline{A}$

A	Y
0	1
1	0



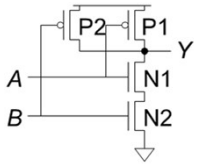
A	P1	N1	*Y*
0	ON	OFF	1
1	OFF	ON	0

CMOS Gates: NAND Gate



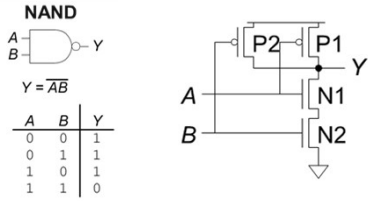
$Y = \overline{AB}$

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0



A	*B*	P1	P2	N1	N2	*Y*
0	0					
0	1					
1	0					
1	1					

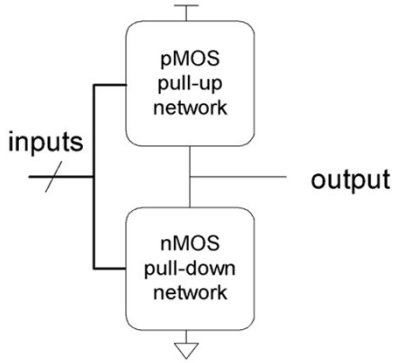
CMOS Gates: NAND Gate



A	*B*	P1	P2	N1	N2	*Y*
0	0	ON	ON	OFF	OFF	1
0	1	ON	OFF	OFF	ON	1
1	0	OFF	ON	ON	OFF	1
1	1	OFF	OFF	ON	ON	0



CMOS Gate Structure





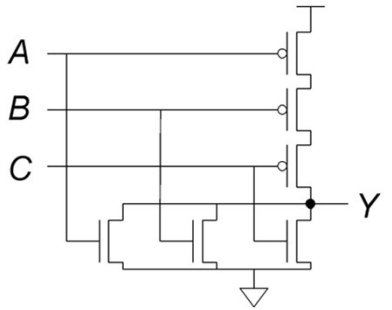
NOR Gate

How do you build a three-input NOR gate?



NOR3 Gate

How do you build a three-input NOR gate?





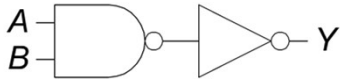
Other CMOS Gates

How do you build a two-input AND gate?



AND2 Gate

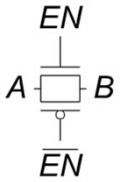
How do you build a two-input AND gate?





Transmission Gates

- nMOS pass 1's poorly
- pMOS pass 0's poorly
- Transmission gate is a better switch
  - passes both 0 and 1 well
- When *EN* = 1, the switch is ON:
  - *EN* = 0 and *A* is connected to *B*
- When *EN* = 0, the switch is OFF:
  - *A* is not connected to *B*





Power Consumption

Power = Energy consumed per unit time

- Dynamic power consumption
- Static power consumption



Dynamic Power Consumption

- **Power to charge transistor gate capacitances**
  - Energy required to charge a capacitance, C, to V_{DD} is CV_{DD}^2
  - Circuit running at frequency f: transistors switch (from 1 to 0 or vice versa) at that frequency
  - Capacitor is charged $f/2$ times per second (discharging from 1 to 0 is free)
- **Dynamic power consumption:**

$$P_{dynamic} = \tfrac{1}{2}CV_{DD}^2 f$$



Static Power Consumption

- Power consumed when no gates are switching
- Caused by the *quiescent supply current*, I_{DD} (also called the *leakage current*)
- Static power consumption:

$$P_{static} = I_{DD}V_{DD}$$



Power Consumption Example

- Estimate the power consumption of a wireless handheld computer
  - V_{DD} = 1.2 V
  - C = 20 nF
  - f = 1 GHz
  - I_{DD} = 20 mA

$$P = \tfrac{1}{2}CV_{DD}^2 f + I_{DD}V_{DD}$$

$$= \tfrac{1}{2}(20\text{ nF})(1.2\text{ V})^2(1\text{ GHz}) + (20\text{ mA})(1.2\text{ V})$$

$$= \mathbf{(14.4 + 0.024)\ W \approx 14.4\ W}$$

